

Exercises

1. User Upcoming Events

Show a list of all upcoming events a user is registered for in their city, sorted by date.

```
SELECT u.full_name, e.title, e.city, e.start_date
FROM Users u
JOIN Registrations r ON u.user_id = r.user_id
JOIN Events e ON r.event_id = e.event_id
WHERE e.status='upcoming'
AND u.city=e.city
ORDER BY e.start_date;
```

full_name	title	city	start_date
Alice Johnson	Tech Innovators Meetup	New York	2025-06-10 10:00:00
Bob Smith	Frontend Development Bootcamp	Los Angeles	2025-07-01 10:00:00

2. Top Rated Events

Identify events with the highest average rating, considering only those that have received at least 10 feedback submissions.

```
SELECT e.title, AVG(f.rating) AS avg_rating
FROM Events e
JOIN Feedback f ON e.event_id=f.event_id
GROUP BY e.event_id,e.title
HAVING COUNT(f.feedback_id)>=10
ORDER BY avg_rating DESC;
```

Empty Set

3. Inactive Users

Retrieve users who have not registered for any events in the last 90 days.

```
SELECT *
FROM Users
WHERE user_id NOT IN
(
SELECT user_id
FROM Registrations
WHERE registration_date>=CURDATE()-INTERVAL 90 DAY
);
```

Empty Set

4. Peak Session Hours

Count how many sessions are scheduled between 10 AM to 12 PM for each event.

```
SELECT e.title,COUNT(s.session_id) AS session_count
FROM Events e
LEFT JOIN Sessions s
ON e.event_id=s.event_id
WHERE TIME(s.start_time)>='10:00:00'
AND TIME(s.start_time)<'12:00:00'
GROUP BY e.event_id,e.title;
```

title	session_count
Tech Innovators Meetup	2
Frontend Development Bootcamp	1

5. Most Active Cities

List the top 5 cities with the highest number of distinct user registrations.

```
SELECT e.city,
COUNT(DISTINCT r.user_id) AS registrations
FROM Events e
JOIN Registrations r
ON e.event_id=r.event_id
GROUP BY e.city
ORDER BY registrations DESC
LIMIT 5;
```

city	registrations
Chicago	2
New York	2
Los Angeles	1

6. Event Resource Summary

Generate a report showing the number of resources (PDFs, images, links) uploaded for each event.

```
SELECT e.title,  
COUNT(r.resource_id) AS total_resources  
FROM Events e  
LEFT JOIN Resources r  
ON e.event_id=r.event_id  
GROUP BY e.event_id,e.title;
```

title	total_resources
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Tech Innovators Meetup	1
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AI & ML Conference	1
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Frontend Development Bootcamp	1
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7. Low Feedback Alerts

List all users who gave feedback with a rating less than 3, along with their comments and associated event names.

```
SELECT u.full_name,e.title,f.comments,f.rating  
FROM Feedback f  
JOIN Users u ON f.user_id=u.user_id  
JOIN Events e ON f.event_id=e.event_id  
WHERE f.rating<3;
```

8. Sessions per Upcoming Event

Display all upcoming events with the count of sessions scheduled for them

```
SELECT e.title,  
COUNT(s.session_id) AS session_count  
FROM Events e  
LEFT JOIN Sessions s  
ON e.event_id=s.event_id  
WHERE e.status='upcoming'  
GROUP BY e.event_id,e.title;
```

title	session_count
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Tech Innovators Meetup	2
------------------------	---

Frontend Development Bootcamp	1
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9. Organizer Event Summary

For each event organizer, show the number of events created and their current status (upcoming, completed, cancelled).

```
SELECT u.full_name,e.status,
COUNT(e.event_id) AS total_events
FROM Users u
JOIN Events e
ON u.user_id=e.organizer_id
GROUP BY u.full_name,e.status;
```

full_name	status	total_events
Alice Johnson	upcoming	1
Bob Smith	upcoming	1
Charlie Lee	completed	1

10. Feedback Gap

Identify events that had registrations but received no feedback at all.

```
SELECT e.title
FROM Events e
JOIN Registrations r
ON e.event_id=r.event_id
LEFT JOIN Feedback f
ON e.event_id=f.event_id
WHERE f.feedback_id IS NULL
GROUP BY e.event_id,e.title;
```

rontend Development Bootcamp

11. Daily New User Count

Find the number of users who registered each day in the last 7 days.

```
SELECT registration_date,  
COUNT(*) AS new_users  
FROM Users  
WHERE registration_date>=CURDATE()-INTERVAL 7 DAY  
GROUP BY registration_date;
```

Empty Set

12. Event with Maximum Sessions

List the event(s) with the highest number of sessions.

```
SELECT e.title,  
COUNT(s.session_id) AS sessions  
FROM Events e  
JOIN Sessions s  
ON e.event_id=s.event_id  
GROUP BY e.event_id,e.title  
HAVING COUNT(s.session_id)=  
(  
SELECT MAX(cnt)  
FROM  
(  
SELECT COUNT(*) cnt  
FROM Sessions  
GROUP BY event_id  
) x  
);
```

title	session s
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Tech Innovators Meetup	2
------------------------	---

13. Average Rating per City

Calculate the average feedback rating of events conducted in each city

```
SELECT e.city,  
AVG(f.rating) AS average_rating  
FROM Events e  
JOIN Feedback f
```

```
ON e.event_id=f.event_id
GROUP BY e.city;
```

city	average_rating
Chicago	4.50
New York	3.00

14. Most Registered Events

List top 3 events based on the total number of user registrations.

```
SELECT e.title,
COUNT(r.registration_id) AS registrations
FROM Events e
JOIN Registrations r
ON e.event_id=r.event_id
GROUP BY e.event_id,e.title
ORDER BY registrations DESC
LIMIT 3;
```

title	registrations
Tech Innovators Meetup	2
AI & ML Conference	2
Frontend Development Bootcamp	1

15. Event Session Time Conflict

Identify overlapping sessions within the same event (i.e., session start and end times that conflict).

```
SELECT s1.event_id,s1.title,s2.title
FROM Sessions s1
JOIN Sessions s2
ON s1.event_id=s2.event_id
AND s1.session_id<s2.session_id
AND s1.start_time<s2.end_time
AND s2.start_time<s1.end_time;
```

Empty Set

16. Unregistered Active Users

Find users who created an account in the last 30 days but haven't registered for any events.

```
SELECT *
FROM Users
WHERE registration_date >= CURDATE() - INTERVAL 30 DAY
AND user_id NOT IN
(
  SELECT user_id
  FROM Registrations
);
```

Empty Set

17. Multi-Session Speakers

Identify speakers who are handling more than one session across all events.

```
SELECT speaker_name,
COUNT(*) AS sessions
FROM Sessions
GROUP BY speaker_name
HAVING COUNT(*) > 1;
```

Empty Set

18. Resource Availability Check

List all events that do not have any resources uploaded.

```
SELECT e.title
FROM Events e
LEFT JOIN Resources r
ON e.event_id = r.event_id
WHERE r.resource_id IS NULL;
```

Empty Set

19. Completed Events with Feedback Summary

For completed events, show total registrations and average feedback rating.

```
SELECT e.title,
```

```

COUNT(DISTINCT r.registration_id) AS registrations,
AVG(f.rating) AS average_rating
FROM Events e
LEFT JOIN Registrations r
ON e.event_id=r.event_id
LEFT JOIN Feedback f
ON e.event_id=f.event_id
WHERE e.status='completed'
GROUP BY e.event_id,e.title;

```

title	registrations	average_rating
AI & ML Conference	2	4.50

20. User Engagement Index

For each user, calculate how many events they attended and how many feedbacks they submitted.

```

SELECT u.full_name,
COUNT(DISTINCT r.event_id) AS events_attended,
COUNT(DISTINCT f.feedback_id) AS feedbacks
FROM Users u
LEFT JOIN Registrations r
ON u.user_id=r.user_id
LEFT JOIN Feedback f
ON u.user_id=f.user_id
GROUP BY u.user_id,u.full_name;

```

full_name	events_attended	feedbacks
Alice Johnson	1	0
Bob Smith	1	1
Charlie Lee	1	1
Diana King	1	1
Ethan Hunt	1	0

21. Top Feedback Providers

List top 5 users who have submitted the most feedback entries

```
SELECT u.full_name,  
COUNT(f.feedback_id) AS feedback_count  
FROM Users u  
JOIN Feedback f  
ON u.user_id=f.user_id  
GROUP BY u.user_id,u.full_name  
ORDER BY feedback_count DESC  
LIMIT 5;
```

full_name	feedback_count
Bob Smith	1
Charlie Lee	1
Diana King	1

22. Duplicate Registrations Check

Detect if a user has been registered more than once for the same event.

```
SELECT user_id,event_id,  
COUNT(*) AS duplicates  
FROM Registrations  
GROUP BY user_id,event_id  
HAVING COUNT(*)>1;
```

Empty Set

23. Registration Trends

Show a month-wise registration count trend over the past 12 months.

```
SELECT DATE_FORMAT(registration_date,'%Y-%m') AS month,  
COUNT(*) AS registrations  
FROM Registrations  
GROUP BY month  
ORDER BY month;
```

month	registrations
2025-04	2
2025-05	2
2025-06	1

24. Average Session Duration per Event

Compute the average duration (in minutes) of sessions in each event.

```
SELECT e.title,
AVG(TIMESTAMPDIFF(MINUTE,s.start_time,s.end_time))
AS avg_duration
FROM Events e
JOIN Sessions s
ON e.event_id=s.event_id
GROUP BY e.event_id,e.title;
```

title	avg_duration
Tech Innovators Meetup	67.5
AI & ML Conference	90
Frontend Development Bootcamp	120

25. Events Without Sessions

List all events that currently have no sessions scheduled under them.

```
SELECT e.title
FROM Events e
LEFT JOIN Sessions s
ON e.event_id=s.event_id
WHERE s.session_id IS NULL;
```

Empty Set